AUTOMATED AEROPONICS LABORATORY SYSTEM

AALS Technologies Innovation

Executive Summary

Development of an automated laboratory system for monitoring and controlling aeroponics agricultural systems using common methods of requirements engineering and analysis in order to gain a better understanding of the optimal product in the current marketplace.

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Introduction: AALS - a new paradigm in the practice of sustainable agriculture

Product Description

Supply chain interruptions during the Covid 19 pandemic have illuminated the need for a product that can provide novice users a means to grow produce in their homes without arable land and with limited resources. A fully automated aeroponics laboratory system provides an accelerated agricultural application for the provision of food. The system tracks data from an array of sensors utilized to log temperature, pH, electroconductivity, oxidation reduction potential and dissolved oxygen. An algorithm based on that data in turn makes decisions to promote growth by adjusting nutrient levels. Timers for lights and nutrient pump operations set a daily schedule for the regular tending necessary to maintain plant health..

Mission Statement

Automated Aeroponics Laboratory System (AALS) is a platform to innovate a micro-farming revolution.

Document Purpose

This document will serve as the AALS Software Requirements Specification (SRS). The AALS SRS is a provision of requirements for Automated Aeroponics Laboratory System an AALS Innovation. An SRS is intended to assist stakeholders in clarifying the proposed functionality of AALS as a product in development. The AALS SRS is a foundational document that will direct development efforts to include production and quality control and will be used as the sole guideline in measuring the effectiveness of the development process. In the event that development efforts are determined to have deviated from the requirements established in the AALS SRS, stakeholders may determine to alter development activities to achieve project-wide conformance to documentation or in turn alter the AALS SRS to adhere to newly established and properly accepted changes.

Document Conventions

Below is a list acceptable acronyms and established conventions identified for the purposes of AALS documentation and implementation of development activities.
SRS: Software Requirements Specification
End User: an individual who has downloaded the AALS controller software for personal use.
Use Case: an abstract diagrammatic representation of AALS controller software.

Intended Audience and Reading Suggestions

The intended audiences for this document are as follows:

AALS End Users
AALS application Stakeholders

A bi-monthly review process has been enacted by AALS Technologies to ensure that AALS projections remain in line with the AALS SRS. Should those projections change over the course

of the project, the SRS will come under review for the purpose of determining the necessity to update the AALS SRS.

Product Development Scope

The AALS system will be developed, validated, and released to production per the directives established in the AALS SRS. The agreed upon benefit to stakeholders is the addition of a revenue stream in the micro-farming market. Consumers who purchase a home aeroponics System will be provisioned with an integrated hardware and software kit. The AALS software component controls the aeroponics system and transfers data to a SLIMS instance with a MariaDB backend.

AALS controller software will provide these functions and modules.

- WiFI Connection
- Dashboard
- Data Logging
- On Display Reporting
- System Calibration
- Data Transfer API
- Automation

Overall Description

The AALS SRS provides an Overall Description and a list of Specific Requirements for the AALS controller software. The Overall Description is an abstract provision of requirements. The Specific Requirements is a detailed provision of requirements.

Product Perspective

AALS controller software is entirely self-contained and accessible from the system-side edge computing hardware module and from a mobile application.

Product Function

AALS controller software consists of two specific end user interfaces. The system-side edge computing hardware module and a mobile application provide access to Data Logging, On Display Reporting, and System Calibration.

AALS controller software system-side edge computing hardware module and mobile application functions are as follows.

Dashboard

- Allows end user to navigate AALS by clicking on buttons
- Shows real-time AALS statistics reflective of sensor array readings

Data Logging Module

- Sends data from sensor array to on-board SD card for storage
- Date, time, and type of reading in the format of a text file to be named with time in

hours_ddmmmyyyy_method of collection
Date, time, and type of exception or error in format of a text file to be named with time in hours_ddmmmyyyy_type of exception/error and to include cause of exception or error and any notation of resolution of exception or error - was it resolved? How was it resolved?

On Display Report Module

Displays the following real time data for the end user:
sensor array readings
menu to navigate to other modules

User Characteristics

Training required for use – negligible
Prior experience using similar apps – negligible
Technical Expertise – negligible

End User Documentation

List of documents available within the AALS software controller system side hardware module and mobile application
Help Index navigation via Dashboard menu
Latest update prompt, if the software requires an update to be pushed by the end user, a notification will pop up when the user opens the mobile application. The end user will not be able to proceed until they authorize the update.

Assumptions and Dependencies
AALS system-side hardware module and mobile application topology

Apache Tomcat web app

Access for client-side interaction

End User access to AALS via Android and IOS or web browser

Key Functionalities

End User may access dashboard, data logging, system calibration, and on display reporting.

Key Classes

Calibrate System – begins calibration script to calibrate sensor array at system-side hardware module.

Show System Status – shows real time readings from sensor array at system-side hardware module and mobile application

Transfer Data - sends flat file to SLIMS and parses data into SLIMS infrastructure

General list of preliminary Use Cases

End User downloads the app from the app store.

End User connects app to system-side hardware module.

End User calibrates AALS.

Preliminary draft of Application Use Case Flow - Calibrate System

End user selects “Calibrate” button in app.

App instructs End user to place each probe in calibration solution.

Calibration script runs.

Readings normalize.

Calibration script completes.

End User selects “Accept Calibration” button.

App displays pop-bubble informing End User that Calibration has been accepted.

App returns to Dashboard.

Preliminary draft of Application Use Case Flow - Show System Status

End user launches system-side hardware module and mobile application

App opens on Dashboard which displays real time system status.

Note - from any other module, End User may select “Dashboard” button from on-screen menu bar to return to the system status screen.

Expanded Use Cases

Use Case 1

End user selects “Calibrate” button in app.

App instructs End user to place each probe in the calibration solution.

Calibration script runs.

Readings normalize.

Calibration script completes.

End User selects "Accept Calibration" button.

App displays pop-bubble informing End User that Calibration has been accepted.

App returns to Dashboard.

1. Name – Calibrate System

2. Brief Description – End user navigates to system calibration screen in system-side hardware module from dashboard by selecting "Calibrate System" button.

3. Actors – End User

4. Basic Flow

- End user opens system-side hardware module to Dashboard screen
- End user selects "Calibrate System" button from Dashboard menu
- Application prompts user to place probes in the calibration solution
- End user selects "Start Calibration" button.
- Automated Calibration script runs
- Application prompts End user to select either "Accept Calibration" or "Discard Calibration" button
- End User selects "Accept Calibration" button
    - Application saves calibration settings to system
    - Application displays message that calibration settings have been saved
- Application returns end user to dashboard

5. Alternate Flows

- End User selects "Discard Calibration" button
    - Application discards calibration settings
    - Application displays message that calibration settings have been discarded

- Application returns End User to Dashboard page

6. Pre-Conditions

   - Sensor Array is successfully integrated to AALS

   - Imbedded Calibration script is present

7. Post-Conditions

   - Accepted Calibration settings have saved successfully and confirmation message displayed in application in pop-up message bubble

   - End user automatically returned to Dashboard

8. Special Requirements

   - If End User navigates to system calibration module, application should display current calibration settings on screen prior to beginning new calibration cycle.

Analysis of Tactics

Identification of Key Stakeholders

There are three groups identified as key stakeholders for the product. AALS Technologies are the founders. They are developers of the product and those that can be described as originators of the intellectual property. These individuals will be majority shareholders in the business associated with this product. Subsequently, the sponsors are identified as those directly supporting integration of the technologies into the marketplace. These individuals will seek to broaden the reach and profit margins of the product. As such, funders hold a substantial stock percentage in the business. Finally, the advisors are involved in guiding the founders toward the first release of the product. The advisors provide feedback which may impact structural revisions and inform the direction of investment by the sponsors in their interest in helping the business to go forward to production.

Identification of Key Drivers

Under Armour purchased MyFitnessPal for $475 Million dollars in 2015. At the time, MyFitnessPal boasted 80 million users. In the first quarter of 2019, MyFitnessPal was identified as having 19.1 million unique users each month and 120 million dollars in revenue growth the prior year. This placed the app strongly as number two in the market. According to a case study by Sharon Lamb, the "Quantified Self" movement is continuously trending upward. Likewise, a research study by Emily Alonso suggests that health and fitness tools with an auxiliary dimension that employs social media interactions are a sound investment in the digital marketplace. Anticipating the success of not just life-logging applications but applications that place the control of health and livelihood squarely in the hands of end users without the need of a professional intermediary. Fully automated micro-farming is a gateway to otherwise highly complex food-producing agricultural applications to individuals with little or no expertise. In two coincidentally overlapping demographics: Bioinformatics Applications Engineers and Chemical Analysis Laboratory Personnel, AALS developers took the opportunity to identify its first customers within this particular cross section of the population. AALS developers chose the following two groups as stakeholders: AALS Technologies and Genohm, a part of Agilent - makers of SLIMS.

Identification of Key Constraints

AALS controller software shall be a system-side hardware module and a mobile phone based system that runs on Android and IOS user interfaces as well as web browsers.
AALS controller software shall be developed using C++ and Java programming languages.
AALS controller software implementation shall delegate automated functionalities to be administered by C++ scripting uploaded directly to the Arduino Due.
*AALS controller software implementation shall conduct continuous deployment activities through container solution via Kubernetes Engine. **Phase II**
AALS controller software client-SLIMS server transactions shall be administered using Java REST API.
*AALS controller software server side implementation shall store data on AWS infrastructure. **Phase II**
AALS controller software implementation shall utilize Tomcat Apache Server.
AALS controller software shall be a low-latency implementation accessible from any geographical location without limitation.

Identification of Key Requirements

System Requirements

A provision of AALS controller software software requirements in detail to be utilized by stakeholders for the purpose of system design and testing. These requirements are depicted as lists, use case diagrams, use case descriptions and the sequence diagrams..

Functional Requirements

Core actions implemented by AALS controller software. These core actions are organized by two interfaces AALS system-side hardware module and mobile application.

Dashboard

The system shall enable the end user to navigate to Calibrate System and Data Logging modules by tapping/clicking dashboard buttons.

Search

The system shall have a search function. The end users shall be able to search flat files of data collected by the sensor array and exception/error reports by date, sensor type, or exception

type.

System Calibration Module

The system shall run an automated calibration script.

The system shall store calibration settings

Report Module

The system shall generate a report based on sensor array data.

The system shall generate a report based on exceptions or errors.

The system shall generate the following reports and store them locally on the SD card.

System Requirements

External Interfaces

User Interfaces

System-side hardware module

Mobile Phone

Hardware Interfaces

System-Side

Arduino Due board

Mobile Application Side

Operating System : minimum Android 10, IOS 13

Memory : 32 GB

Software Interfaces

Web Server

Tomcat Apache Server version 2.4.4 or later

Programming Languages

C++

Java

Compatible with multiple UI platforms

Android, IOS compatibility

Compatibility must match most current Android and IOS releases beginning with Android 10.0 and IOS 13.

Backwards compatible with five generations of Android and IOS releases, beginning with Android 10.0 and IOS 13.

Software compatibility check must be run at download. Failures will prevent installation of software.

Compatible with multiple hardware types - memory requirements

Minimum storage 128/512 GB

Minimum RAM 16 GB

MicroSD up to 1tb

Battery minimum 5,000 mAH battery

Minimum CPU Qualcomm Snapdragon 865 or Exynos 990 for Android and Apple A13 Bionic for IOS

Software specific phone settings must be activated. Failure to activate these settings will prevent installation of software.

Secondary authentication activated, i.e. authentication via biometrics or security pin. Security check failure will prevent installation of software.

Phone must allow push notifications from software..

Phone must allow use of email, internet access.

Communications Interfaces

4G LTE or higher

All transactions initiated between Mobile Application and Google Cloud must implement Secure Socket Layer (SSL) technology to encrypt the data.

Non-Functional requirements

Nonfunctional requirements include performance, reliability, safety, security and design constraint requirements.

Performance Requirements

The system response time must be less than 200 ms.

The system must process the number of transactions per ms for historical logging data.

Reliability Requirements

5.2.1 AALS controller software shall have a minimum uptime of 99 % excluding time pre-scheduled for maintenance and/or upgrades.

User Requirements

Software must provide an Ad-Free user experience

Software User Interface must be simple and easy enough to navigate without instruction by novice at time of download.

Software must be available for download in official Android and IOS stores. Download page must provide a basic description of the app, developer contact information, verified download stats, and end user reviews.

End users must "e-sign" an End User license agreement via clicking on a check box.

End users must "e-sign" a waiver of liability using a signature box and employing the use of their finger to sign their name via the touchscreen of their mobile device.

Safety Requirements

5.3.1 All AALS controller software data must be backed up every 24 hours and the backup copies stored by Antsle.

Security Requirements

All communication between system-side hardware module SLIMS server side instance for AWS implementations shall be encrypted by using Secure Socket Layer (SSL) technology.

Transaction initiation between AALS server-side hardware module and SLIMS server side instance is only applicable for flat file data transfers. These transactions shall be done using Secure Socket Layer (SSL) technology to encrypt the data.

Access to phone widgets without typing in passcode must be disabled on the phone.

Phone passcode must be enabled and meet the following complexity requirements - alphanumeric with at least one special character and a minimum of eight characters

Phone location services must be activated

Security software which employs Ad blocking and Malware prevention must be present and enabled on phone

Phone cannot be "jail-breaked" or unlocked. Security check results that determine the phone is unlocked or jail-breaked will result in a failure to install the software.

Quality Requirements

AALS controller software source code shall be process documented for the purpose of continued initiatives involving maintenance and upgrade.

Exceptions or errors logged by the system must be categorized for optimized filter-based searching and stored for 12 months on local SD card.

Load time must not exceed 200 ms

Response time must not exceed 200 ms

Maximum Memory Usage not to exceed 250 kb

User navigation must be tracked, categorized for optimized filter-based searching and stored for 12 months on local SD card

Software must check for updates once every thirty days.

Internal system software infrastructure must be in place for automated continuous deployment of updates to the latest release.

98% unit test coverage for front and backend code

Acceptance tests for all scenarios identified during elicitation process

Periodically reviewed core of security checks and vulnerability tests implemented during quality control. Review must continuously expand these core checks and tests to accommodate the changing landscape of mobile apps based on newly discovered vulnerabilities.

Business Requirements

Business requirements for the AALS controller software have been organized in this document as a preliminary statement of customer and business benefit.

Pertinent Questions:

What problem is being solved?

AALS controller software provides a means to control and automate aeroponics agricultural applications.

Why is the problem being solved?

Food deserts and food shortages have become an ever more serious consideration in the modern world. Micro-farming is a means for every person to provide themselves access to fresh produce without in-depth knowledge of how to farm.

Benefits of a successful solution?

It is a competitive market and similarly designed products have seen quite a bit of success. However, in the case of AALS, this is a unique solution demanded by a unique intersection of modern circumstances. The addition of an AI algorithm for the purpose of decision-based automation will be the first implementation of this particular design. Success of this solution will capture a diverse index of publicity stemming from it's uniqueness which will lead to a greater percentage of adoption.

Metrics that determine a successful solution?

AALS purchase statistics

ratings from end users

number of support requests made by users each month

ratings from domain experts

Potential return on investment of a successful solution?

Percent profit margin

Number of unique AALS purchases per month

Cost of acquisition upon corporate purchase

Cost of IPO

Activities included in a successful solution? Activities excluded in a successful solution?

Toolkits for specific types of agricultural implementation

Platform partnerships with major agricultural distributors

Schedule new releases to coincide with new partnership announcements

Business Requirements from the perspective of Business rules:

AALS controller software business requirements are being developed prioritizing toolkit purchases which will be used to drive profit. Upon confirmation of return on investment, profit will be sought through expansion to AALS controller software branded marketing items such as pre-made nutrient solutions, nursery kits, and light bulbs. Customer support interactions associated with use of the mobile app will be leveraged as user acceptance testing to guide refactoring plans for subsequent releases.

Requirements from Facts

Every instance of AALS controller software as determined by end user login must have a unique AALS system.

Every instance of AALS controller software that is installed must have a build number that corresponds with a certificate of software quality provided by the Manufacturing Quality Control division.

Requirements from Constraints

AALS controller software client-server implementation must utilize Java API.

AALS controller software client-server implementation must utilize Android and IOS user interface architecture.

AALS controller software client-server implementation must operate as both a system-side hardware module as well as a mobile phone solution.

Discussion of Applicability

Requirements Elicitation Process

During the development of AALS controller software, five Elicitation Processes were implemented. The first of these was a session dedicated to brainstorming. Brainstorming is a collective effort to identify user stories for a backlog. Each participant offers their best guess for an appropriate story based on shareholder expectations. Brainstorming is not an appropriate activity for a teleconference or Webex for organizational purposes. It is easier to conduct such a session in person to better mitigate issues associated with people talking at the same time. Brainstorming for AALS controller software was completed as an informal first step conducted over visits to local agricultural product manufacturers and distributors. These visits were conducted as a means to help formulate ideas for the product.

Preliminary information gathering activity was completed during an Ethnographic Observation performed at SLIMS offices located at 334 Blackwell St. Suite B007 in Durham, NC. Ethnographic Observation is an effort by a software engineer to gather information about the domain by observing the customer or domain experts on site. This effort can not be conducted remotely because it must be a direct on-site observation by definition. These observations took place in early 2020 the week of February 3rd in New Jersey and February 10th in Durham. The outcome of this research is a full accounting of the domain including those details that can be obtained only through having witnessed them first hand.

Post Ethnographic observation, the Elicitation process continued with a series of interviews. These interviews were initiated as an investigative effort to gather information through interactions with customers and domain experts. Developers devised a selection of questions with the intention that the answers yield substantive outcomes. Interviews were conducted via Slack messaging and Webex teleconferences. Key interviewees in this process were Marcia Murto, SLIMS Project and Business Process Manager, Lisa Buurma, SLIMS Project Manager, and Stien Rooman, SLIMS Development Engineer.

Next, a domain Analysis was performed in an effort by AALS controller software engineers to research the area of expertise upon which the product is dependent or to which the product pertains. Domain Analysis was conducted via the Samuel C. Williams Library at Stevens University and through queries using Google search engine. A selection of articles used to reach conclusions for analysis of AALS controller software domain are provided in the resource list for this paper.

A Task Analysis was initiated in order to supplement analysis of the domain. A task analysis is a decomposition of anticipated relevant tasks that will be implemented by the system. This means that at each step during this particular elicitation process the definition of the task becomes progressively less abstract and more granular in terms of detail. Task Analysis can provide a complete prototype for the product built on the infrastructure provided by domain analysis, interviews, ethnographic observation and brainstorms. This process was conducted remotely as

participants were not co-located at the time of implementation. Task Analysis meetings took place during weekly one hour long Webex teleconferences with the Marcia Murto, SLIMS Project and Business Process Manager, Lisa Buurma, SLIMS Project Manager, and Stien Rooman, SLIMS Development Engineer.

Finally, scenarios were designed as a conclusion to elicitation process activities. Scenarios are tools implemented in behavior driven development. Scenario design is intended to be as detailed as possible and are an ideal means of testing iterative prototypes garnered from task analysis efforts. Scenarios are most effectively implemented in an effort to design as authentic a user experience as possible from clearly defined requirements. Scenarios design took place remotely during weekly Webex teleconferences intended to derive maximum insight from the selected domain experts. Successful scenario design during AALS controller software elicitation was predicated on the effort of AALS controller software engineers to reach adequate depth in their investigative approach as they are constructing scenarios.

Conclusion

Requirement Validation and Analysis

Identification of a successful means of implementing Requirement Validation and Analysis was pursued via research conducted upon Genohm, a part of Agilent - Makers of SLIMS. Genohm is a successful start-up acquired by Agilent in May of 2018. Originally, Genohm was a two-person bioinformatics company founded in 2002. SLIMS, Genohm's core bioinformatics software product, was developed over the course of seven years. Genohm uses a Jira ticketing system and workflow that proceed from opening a ticket to closing a ticket via a review process which incorporates multiple assignees for validation and analysis.

AALS controller software implements a Jira ticketing system similar to that utilized by Genohm. A ticket is entered by a "Reporter" as an issue for the project. Each issue is created with a designation which may be chosen from a list. The issue may be categorized as: Story, Epic, User Need, Defect, Cross-Team Dependency, test Plan, Test Set, Build, Task, Test, or Change request. Under a tab designated "General", the "Reporter" fills out detailed information for the issue. General information includes: a summarizing title, detailed description, Assignee, Status (as determined by the current workflow step), Priority, Fix Versions, Verified in Build/s, Environment (software platform), Attachment (may be dragged and dropped into the field), Labels (for filter searches), Topology, Hours for set-up, and hours for execution. Most issues include tasks that are intended to be completed during the in development step of the workflow. These tasks are identified in the detailed description field. Once all of the fields on the "General" tab have been declared, the "Reporter" may submit the ticket.

Following ticket submission, designated assignees receive an email notification encouraging them to review the ticket. Workflow for the ticket proceeds from ready for development, to in development, to waiting for verification, to testing, and finally to closed. If an assignee identifies a problem requiring resolution at any step in the workflow, a block request may be submitted. Block requests will be approved or denied after review. An approved block request will send the ticket to investigation. Otherwise, if the block request is denied, the ticket proceeds along the workflow pathway. An investigation of a block will proceed with further review and conclude with a request that will send the issue back to the beginning of the workflow with new, edited or additional development tasks.

Resource List

List of "Users"

AALS Technologies

Additional Resources

Case Study Analysis of MyFitnessPal and its Role in the 'Quantified Self' Movement by Sharon Lamb - 11343011
https://quantifiedselfhealth.weebly.com/nutrition--fitness.html

CNN Business Stock Price Forecast for Under Armour
https://money.cnn.com/quote/forecast/forecast.html?symb=UA

MyFitnessPal: How This Winner-Took-All by Helping People Lose
Emily Alonso UC Davis March 20, 2015
Berkeley Roundtable on the International Economy
https://brie.berkeley.edu/sites/default/files/myfitnesspal-how-this-winner-took-all-by-helping-people-lose1.pdf

Software Requirements Specification Example, ABC Online
Movie Booking System
ABC Online Movie Booking System (OMBS)
Version 1.0
Prepared by probyte2u
November 6, 2010

Under Armour CEO: Connected fitness unit logs $120 million in 2018 revenue, brings consumer insights
Larry Dignan for Between the Lines February 13, 2019 ZDNet
https://www.zdnet.com/article/under-armour-ceo-connected-fitness-unit-logs-120-million-in-2018-revenue-brings-consumer-insights/

Software Development Jira Workflow SID v4.07, Genohm, A Part of Agilent - Makers of SLIMS

A fuzzy micro-climate controller for small indoor aeroponics systems. Bambang Dwi Argo*, Yusuf Hendrawan, Ubaidillah Ubaidillah. TELKOMNIKA, Vol.17, No.6, December 2019, pp.3019~3026
https://www.researchgate.net/profile/Bambang_Argo/publication/337976447_A_fuzzy_micro-climate_controller_for_small_indoor_aeroponics_systems/links/5e468a9d299bf1cdb928d9a5/A-fuzzy-micro-climate-controller-for-small-indoor-aeroponics-systems.pdf

Evaluation of the growth, photosynthetic characteristics, antioxidant capacity, biomass yield and quality of tomato using aeroponics, hydroponics and porous tube-vermiculite systems in bio-regenerative life support systems. Wang Minjuan, Dong Chen, Gao Wanlin. Life Sciences in Space Research 22 (2019) 68-75.
https://www.sciencedirect.com/science/article/abs/pii/S221455241930001X

Interviews:
SLIMS Team, Genohm, A Part of Agilent - Makers of SLIMS

Websites
https://northernhomestead.com/high-pressure-aeroponics-guide/
https://aeroponicsdiy.com/
https://www.youtube.com/watch?v=jAUv5YxIkS4
https://www.arduino.cc/en/Tutorial/HomePage